1. Which of the following statements regarding LSTM and Recurrent Neural Network gradients are true?

[] The gradient of the loss with respect to the hiddens is: $\partial L/\partial h(t)$ = ... (Complex formula involving U^T, V^T, R^T)

[] Because of the simple structure of the memory cell, the LSTM architecture fails to be Turing complete.

[] The memory cell fulfills $\partial c(t)/\partial c(t-\tau) = I$ (neglecting dependencies via gates) for $\tau \in \{1, ... t-1\}$. This solves the vanishing gradient problem and is called constant error carousel.

[] The LSTM architecture has no exploding or vanishing gradients because $\partial h(t)/\partial h(t-1)$ is a diagonal matrix.

[] If we adapt $z(t) = \sigma(Ux(t) + Ph(t-1))$, then $z(t) \geq 0$ for all t and the memory cells will always increase. This can be a problem for very long sequences but helpful for certain problems.

2. What are the benefits of a Mixture-of-Experts (MoE) architecture in modern LLMs?

[] It allows for a massive increase in total parameter count (model capacity) without a proportional increase in inference cost.

[] It reduces the total storage size of the model on disk compared to a dense model.

[] It uses sparse activation, where a router selects only a small subset of experts for each token.

[] It activates every single parameter for every token to ensure maximum accuracy.

3. What are the primary functions of the gates in a standard LSTM?

[] The Output Gate scales the input vector before it enters the cell.

[] The Forget Gate allows the memory cell to reset itself when information is no longer needed.

[] The Input Gate protects the memory cell from irrelevant inputs.

[] The Forget Gate determines the final hidden state output to the next layer.

4. Regarding the Encoder-Decoder architecture for sequence learning:

[] The encoder compresses the variable-length input sequence into a fixed-length vector representation.

[] It allows mapping input sequences to output sequences of different lengths (e.g., translation).

[] The input and output sequences must always have the exact same length.

[] It can only be implemented using Transformer networks, not RNNs.

5. Given input vectors x(t), hidden pre-activation s(t), parameter matrices R, W, V... Which of the following statements are true?

[] The gradient of the loss with respect to the recurrent weights R can be written as $\partial L/\partial R$ = sum(δ(t) a^T(t-1)).

[] The deltas fulfill the recursive relation $\delta(t)$ = diag(f'(s(t))) (V^T $\partial L(t)/\partial y(t)$ + R^T $\delta(t-1)$).

[] BPTT is a common regularization technique for recurrent neural networks.

[] The asymptotic complexity of BPTT is O(T^2).

[] The gradient of the loss with respect to the input weights W can be written as $\partial L/\partial W$ = sum(δ(t) x^T(t)).

6. How are sequences typically generated using autoregressive models (like LSTMs or GPTs)?

[] The model predicts a probability distribution for the next token given the history.

[] The model requires future tokens to generate the current token.

[] The entire sequence is generated in parallel in a single shot.

[] The predicted token is fed back into the model as part of the input for the next time step.

7. Which of the following statements is true regarding the memory of mLSTM?

[] The size of the memory matrix C ∈ R^(dxd) is dependent on the sequence length L.

[] The memory matrix C ∈ R^(dxd) is independent of sequence length L and has a capacity dependent on d.

[] The size of the memory matrix is constant and depends on W_k.

[] The matrix memory is computed by taking the old matrix memory and adding a covariance update rule using v_k k_k^T.

[] The size of the matrix memory of mLSTM grows linearly with the length of the input sequence L.

8. Comparing xLSTM and Transformers based on recent benchmarks:

[] xLSTM offers linear computation complexity with respect to sequence length.

[] xLSTM outperforms Transformers on certain state tracking and formal language tasks (e.g., Parity, Dyck languages).

[] xLSTM relies entirely on dot-product attention for mixing information.

[] xLSTM struggles with rare tokens compared to Transformers.

9. Which of the following statements about RNNs are true?

[] An RNN is equivalent to a feedforward network applied to the current sequence element and an internal state representation that depends on earlier sequence elements.

[] Many RNN architectures are Turing complete. This means that they can be trained to exactly imitate the behavior of any computer program.

[] A disadvantage of RNNs is that they can only be applied to sequences of a fixed length.

[] One advantage of RNNs is that their computations can be easily parallelized over time steps.

[] The number of RNN parameters is independent of the sequence length.

10. Which of the following statements regarding the Mamba architecture are correct?

[] It employs a Parallel Scan algorithm (associative scan) to speed up training, avoiding sequential bottlenecks.

[] It utilizes a Selection Mechanism that makes the SSM parameters (B, C, Delta) dependent on the input.

[] It relies mainly on Multi-Head Self-Attention layers like the Transformer.

[] Its inference complexity scales quadratically with sequence length (O(L^2)).

11. Which statements about Bidirectional RNNs/LSTMs are true?

[] They consist of two layers processing the sequence in opposite directions.

[] They are primarily used for real-time time-series forecasting.

[] They are suitable for tasks where the entire sequence is available (e.g., translation), but not for real-time online prediction.

[] They reduce the parameter count by half compared to standard RNNs.

12. Which statements about LSTM dropout are correct?

[] LSTM needs special dropout techniques because a naive application of dropout will result in vanishing/exploding gradients.

[] Dropout is a regularization technique that randomly sets activations to zero during the forward pass.

[] It is especially important to apply dropout when evaluating the model.

[] The central idea of zoneout (Krueger et al., 2017) is to randomly clear the inputs x(t) instead of the memory cell c(t).

[] LSTM needs special dropout techniques because with a naive application the probability of a memory cell to 'survive' decreases exponentially as the sequence grows in time.

13. How do RNNs and LSTMs relate to the vanishing gradient problem?

[] LSTMs mitigate this via the Constant Error Carousel (CEC), which enforces a local derivative of 1 for the cell state.

[] In standard RNNs, gradients can vanish because the error signal is repeatedly multiplied by the weight matrix W during backpropagation.

[] The LSTM forget gate was introduced specifically to cause vanishing gradients for irrelevant data.

[] Standard RNNs naturally avoid vanishing gradients due to their recurrent structure.

14. Which statements about LSTM are true?

[] 'Ticker steps' refers to the technique of letting an LSTM run for more time steps even after the input sequence has been taken in by the network.

[] To counter the drift effect, the input gate should be initialized with positive values.

[] The inputs to the lightweight LSTM have to fulfill the Markov property.

[] It is often beneficial to initialize the forget gate with positive bias values.

[] The focused LSTM network feeds the input vector x(t) only into input gate i(t) and output gate o(t) but not into the block input z(t).

[] The forget gate reintroduces the vanishing gradient problem to a certain degree.

[] An LSTM with coupled input gate and forget gate, i.e., c(t) = (1 - i(t)) * c(t-1) + i(t) * z(t), leads to the gated recurrent unit (GRU).

[] Most LSTM networks used in practice do not feature a forget gate.

15. Which of the following statements about modern open-weight LLM architectural design are correct?

[] DeepSeek V3 uses Multi-Head Latent Attention instead of standard MHA; this compresses key and value tensors to reduce KV cache.

[] Grouped-Query Attention reduces memory usage during inference by sharing key/value projections among several query heads.

[] OLMo 2's architectural novelty compared to other LLMs mainly lies in adopting Mixture-of-Experts layers.

[] Gemma 2 uses sliding window attention as a computational efficiency trick that restricts context scope locally.

[] The original transformer's Post-Norm placement of normalization layers is used unchanged in most modern open LLMs.

[] Multi-Head Latent Attention is functionally equivalent to standard Mixture-of-Experts.

16. Comparing Backpropagation Through Time (BPTT) and Real-Time Recurrent Learning (RTRL):

[] BPTT can be performed in real-time without storing past states.

[] RTRL is significantly faster than BPTT for large networks.

[] RTRL does not need to store sequence history ('local in time') but has a high computational complexity (O(N^4)).

[] BPTT requires storing activations for the entire sequence history, making it memory intensive.

17. Why does SmolLM utilize 'NoPE' (No Positional Embeddings)?

[] NoPE requires the model to use RNN layers for position tracking.

[] It was found that NoPE generalizes better to sequence lengths longer than those seen during training (length extrapolation).

[] It relies on the causal attention mask to provide an implicit sense of token order.

[] NoPE is used because it reduces the training time by 50%.

18. In the standard Scaled Dot-Product Attention equation, which statements are true?

[] The term QK^T computes a similarity score between Queries and Keys.

[] The result of the softmax is a single scalar value.

[] The formula is Attention(Q, K, V) = softmax(QK^T / sqrt(d_k))V

[] The softmax is applied to the Values V directly.

[] The formula is Attention(Q, K, V) = softmax(QK^T / sqrt(d_k)) / V

[] The scaling factor 1/sqrt(d_k) is used to prevent the dot products from becoming too large, avoiding softmax saturation.